Hybrid guessing depth, Curve25519 miniature

Curve25519 miniature: hybrid guessing depth (bits and isogeny forms)

Protocol: PROTOCOL.md (frozen, with Amendment 1). Scope of every statement below: **k = 20, m = 2, mode full, lifts 0, 100 random subgroup targets per cell, only the G values in the frozen grid**. Nothing here is claimed for other k, m, or G outside the grid.

What was built

`pkm-semaev guess=G` now also applies to the isogeny membership form (code in `src/cryptanalysis/ic_framework/pkm_semaev.rs`): the first `t-G` Velu maps are applied to every element of `V`, `V` is grouped into `2^G` groups of `2^(t-G)` by the value reached at level `t-G`, and one branch system per group is solved (summand 1 on the sub-tower, the other unchanged). `G >= t` is refused. Bits guessing is unchanged. The code compiled on the first try; no bug was found in it.

Test `isogeny_guessing_agrees_with_grid` (new, in the WIP commit): isogeny form, t = 4, `G in {1, 2}`, through the shared helper `agrees_with_grid`: F4 verdict == grid verdict == exact decomposable set on 32 random/planted targets, relation sums to the target, zero crosscheck disagreements, F4 work counted; the `2^G` groups have `2^(t-G)` elements each (checked for G = 2); `guess=t` is refused.

Summary line of `cargo test --release --lib pkm_semaev`:

```
test result: ok. 11 passed; 0 failed; 0 ignored; 0 measured; 6075 filtered out; finished in 536.14s
```

(10 pre-existing tests matching the filter plus the new one.) `cargo clippy --release --lib --bin curve25519_family -- -D warnings` exits 0 (only the environmental unknown-lint warning). rustfmt was applied afterwards (whitespace only, 10 lines); the tests ran on the pre-rustfmt source, the frozen binary was built from the formatted source.

Measurement

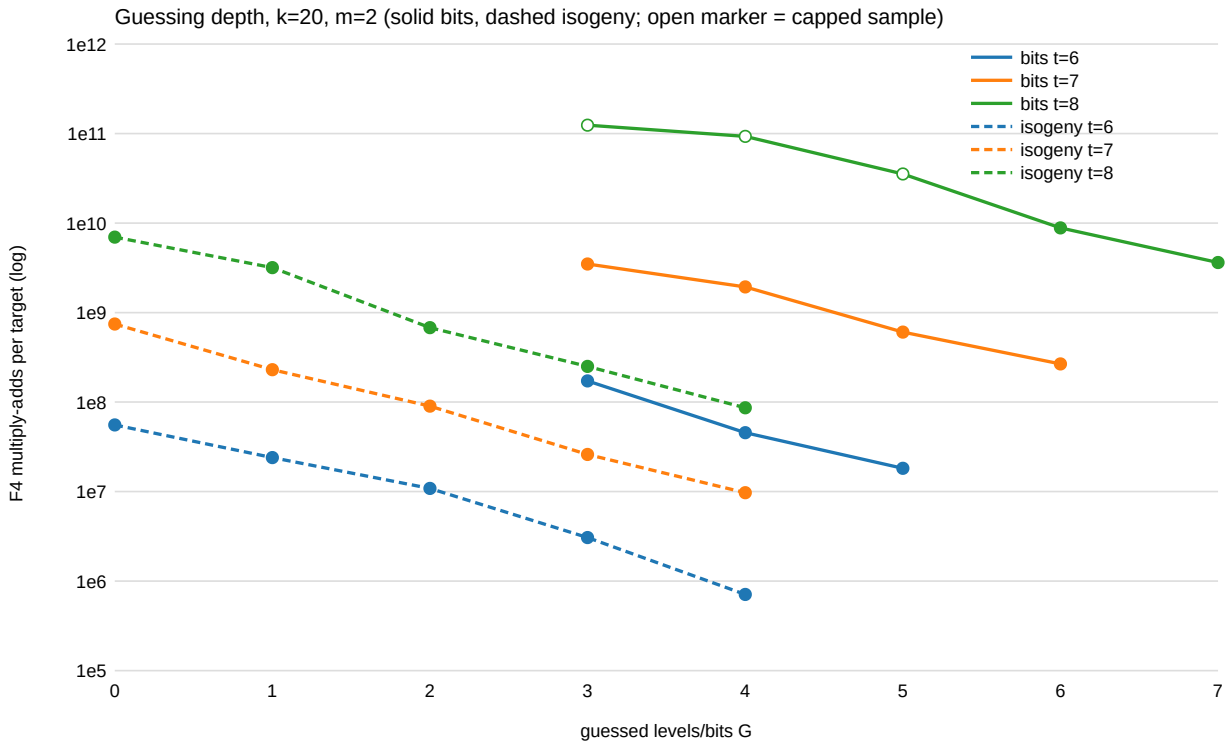
Binary `/home/user/frozen-bin/curve25519_family_lead4`, sha256 `55f44f48dcac5b54792aab59eda5edd7bae8a45ff27df9f0714322245224b0af` (`analysis/binary.sha256`). Driver `analysis/run_cells.sh` (shell + jq), each cell under `ulimit -v 3500000` and `timeout 900`, sequential, seed `2026111400+t`, cap 600 s. Raw: `analysis/f4cost_guess.jsonl`, status `analysis/f4cost_guess.jsonl.status` (27 cells, 27 ok, 0 failed, all stderr files empty and not committed), tables by `analysis/make_tables.sh` (jq + awk). Total wall 56 min of F4 time; the whole grid ran. Run locally, not on EKS (a development measurement inside the lead; no L2 or cross-method claim).

Columns: mul/tgt = F4 multiply-adds per target (all branches charged); calls/tgt; C_solve = multiply-adds per F4 call; C(G) = 2^G x C_solve; deg = max solving degree; und = undecided systems; h/v = hits / verified hits; PR(lam) = mul/tgt / lambda (frozen lambda from the protocol); cov = harness `coverage_exact`; PR(cov) = mul/tgt / cov (Amendment 1). CAP = wall cap hit, only the first n targets done (sample truncated).

form	t	G	targets	cap	mul/tgt	calls/tgt	C_solve	C(G)	deg	und	h/v	PR(lam)	cov	PR(cov)
bits	6	3	100/100	-	1.72e8	8.0	2.16e7	1.72e8	6	0	0/0	1.81e10	0.001234	1.4e11
bits	6	4	100/100	-	4.55e7	16.0	2.85e6	4.55e7	5	0	0/0	4.78e9	0.001234	3.69e10
bits	6	5	100/100	-	1.82e7	32.0	5.69e5	1.82e7	5	0	0/0	1.91e9	0.001234	1.47e10
bits	7	3	100/100	-	3.48e9	8.0	4.36e8	3.48e9	6	0	0/0	6.98e10	0.006280	5.55e11

form	t	G	targets	cap	mul/tgt	calls/tgt	C_solve	C(G)	deg	und	h/v	PR(lam)	cov	PR(cov)
bits	7	4	100/100	-	1.93e9	16.0	1.21e8	1.93e9	6	0	0/0	3.88e10	0.006280	3.08e11
bits	7	5	100/100	-	6.04e8	32.0	1.89e7	6.04e8	6	0	0/0	1.21e10	0.006280	9.62e10
bits	7	6	100/100	-	2.67e8	64.0	4.18e6	2.67e8	6	0	0/0	5.36e9	0.006280	4.26e10
bits	8	3	32/100	CAP	1.24e11	7.875	1.58e10	1.26e11	7	0	1/1	6.91e11	0.024740	5.02e12
bits	8	4	40/100	CAP	9.31e10	15.8	5.89e9	9.43e10	7	0	1/1	5.19e11	0.024740	3.76e12
bits	8	5	75/100	CAP	3.54e10	31.453333333333333	1.12e9	3.6e10	7	0	2/2	1.97e11	0.024740	1.43e12
bits	8	6	100/100	-	8.82e9	62.73	1.41e8	9e9	6	0	3/3	4.92e10	0.024740	3.57e11
bits	8	7	100/100	-	3.63e9	125.45	2.89e7	3.7e9	6	0	3/3	2.02e10	0.024740	1.47e11
isogeny	6	0	100/100	-	5.55e7	1.0	5.55e7	5.55e7	5	0	0/0	5.83e9	0.001310	4.23e10
isogeny	6	1	100/100	-	2.4e7	2.0	1.2e7	2.4e7	5	0	0/0	2.52e9	0.001310	1.83e10
isogeny	6	2	100/100	-	1.08e7	4.0	2.71e6	1.08e7	5	0	0/0	1.14e9	0.001310	8.27e9
isogeny	6	3	100/100	-	3.07e6	8.0	3.83e5	3.07e6	5	0	0/0	3.22e8	0.001310	2.34e9
isogeny	6	4	100/100	-	7.1e5	16.0	4.44e4	7.1e5	5	0	0/0	7.45e7	0.001310	5.42e8
isogeny	7	0	100/100	-	7.46e8	1.0	7.46e8	7.46e8	5	0	0/0	1.19e10	0.009054	8.23e10
isogeny	7	1	100/100	-	2.3e8	2.0	1.15e8	2.3e8	5	0	0/0	3.67e9	0.009054	2.54e10
isogeny	7	2	100/100	-	9.03e7	4.0	2.26e7	9.03e7	5	0	0/0	1.44e9	0.009054	9.97e9
isogeny	7	3	100/100	-	2.6e7	8.0	3.24e6	2.6e7	5	0	0/0	4.14e8	0.009054	2.87e9
isogeny	7	4	100/100	-	9.7e6	16.0	6.06e5	9.7e6	5	0	0/0	1.55e8	0.009054	1.07e9
isogeny	8	0	100/100	-	6.98e9	1.0	6.98e9	6.98e9	5	0	4/4	2.83e10	0.035121	1.99e11
isogeny	8	1	100/100	-	3.18e9	1.98	1.6e9	3.21e9	5	0	4/4	1.29e10	0.035121	9.04e10
isogeny	8	2	100/100	-	6.8e8	3.91	1.74e8	6.96e8	5	0	4/4	2.76e9	0.035121	1.94e10
isogeny	8	3	100/100	-	2.5e8	7.8	3.2e7	2.56e8	5	0	4/4	1.02e9	0.035121	7.12e9
isogeny	8	4	100/100	-	8.64e7	15.54	5.56e6	8.89e7	5	0	4/4	3.51e8	0.035121	2.46e9

(`C(G)` differs from mul/tgt only where branches did not all run to a full `2^G` per target: the capped bits t = 8 cells and isogeny t = 8, G = 1..4, where a target that hits stops before running all 2^G branches; this is the reading consistent with calls/tgt < 2^G, not separately verified.)



muladds per target vs G

`analysis/guess.svg`: multiply-adds per target vs G, log scale, per (form, t); open markers are capped samples.

Argmin G per (form, t) (muladds per target, ties to smaller G)

form	t	argmin G	grid range	at the grid edge?
bits	6	5	3..5	yes (G = t-1)
bits	7	6	3..6	yes (G = t-1)
bits	8	7	3..7	yes (G = t-1)
isogeny	6	4	0..4	yes (grid max)
isogeny	7	4	0..4	yes (grid max)
isogeny	8	4	0..4	yes (grid max)

Every cost series is strictly decreasing over the whole measured G range; no minimum is bracketed. The “argmin” is therefore the edge of the grid, not an optimum: the true optimum is at or beyond it for isogeny, and for bits it is at the largest allowed G (G < t) as far as measured.

Predictions

- **P1 (bits G* = 3, 4, 5 for t = 6, 7, 8): false.** Measured argmin 5, 6, 7 (edge). Cost falls at every step: t = 6: 1.72e8, 4.55e7, 1.82e7; t = 7: 3.48e9, 1.93e9, 6.04e8, 2.67e8; t = 8: 1.24e11 (capped), 9.31e10 (capped), 3.54e10 (capped), 8.82e9, 3.63e9. The t = 8 G = 3..5 cells are truncated samples (32/40/75 targets) and are only inconclusive as to their own level; the ordering of the uncapped G = 6 and 7 cells, and the large gaps to G = 3..5, do not depend on it. The “optimum rises by about one per unit of t” shape is not seen; the optimum is not below G = t-1.
- **P2 (isogeny G* = 1, 2, 3): false.** Measured argmin 4, 4, 4 (grid max) at all three t. The optimum is not bracketed; it is at least 4.
- **P3 (isogeny + guessing beats bits + guessing per relation at equal t): true on both bases.** Best isogeny vs best bits cell, PR(lam): t = 6: 7.45e7 vs 1.91e9; t = 7: 1.55e8 vs 5.36e9; t = 8: 3.51e8 vs 2.02e10. PR(cov): t = 6: 5.42e8 vs 1.47e10; t = 7: 1.07e9 vs 4.26e10; t = 8: 2.46e9 vs 1.47e11. Ratios 26x to 58x (lam) and 27x to 60x (cov). It also holds at equal G (G = 3, 4 where both forms have cells) at every t.
- **P4 (best per-relation cost is isogeny at t = 7 or 8 with its optimum G, below the previous best): false as stated, on both bases (they agree).** The lowest per-relation cost in the grid is isogeny t = 6, G = 4: 7.45e7 (lam) / 5.42e8 (cov), against the t = 7 and t = 8 isogeny minima (G = 4) of 1.55e8 / 1.07e9 and 3.51e8 / 2.46e9. All three are below the previous best (isogeny t = 6, G = 0: measured here 5.83e9 (lam) / 4.23e10 (cov), consistent with the protocol’s 5.9e9 and 4.2e10): the t = 6 cell is 78x (lam) and 78x (cov) lower. Per-relation cost rises with t at the grid-max G for isogeny. Since the t = 6 optimum is itself at the grid edge, “best overall” is the best measured cell, not a proven minimum.
- **P5 (C(G) is U-shaped): not observed in the grid.** C(G) falls monotonically in all six series (for the uncapped series, e.g. isogeny t = 7: 7.46e8, 2.30e8, 9.03e7, 2.60e7, 9.70e6). The solving degree is flat or falling (bits 7 -> 6 at t = 8 as G rises, 6/5/5 at t = 6; isogeny 5 throughout), so no point where the 2^G factor dominates was reached. The right side of a U, if it exists, lies beyond the measured G.

Verified hits equal hits in every cell (bits t = 8: 1, 1, 2, 3, 3; isogeny t = 8: 4 in every cell; all other cells 0 hits of 100 targets, as expected from coverage 0.001 to 0.009). Undecided = 0 in every cell. At 0-4 hits per cell the hit counts say nothing about yield beyond consistency with `coverage_exact` (bits t = 8: 0.025, isogeny t = 8: 0.035, i.e. 2.5 and 3.5 expected per 100).

Reading: the lam and cov columns differ by a constant factor per (form, t) (the Amendment 1 cofactor effect); this does not change any ordering above.

Deviations and caveats

- None in cells, seeds, or rules; no data-dependent change after the first cell.

- Three bits t = 8 cells (G = 3, 4, 5) hit the 600 s cap (615 s, 604 s, 602 s wall) and report 32, 40, 75 of 100 targets. Their per-target means are on truncated samples. Protocol classification: inconclusive as to their own absolute level.
- The grid ends before any minimum: the stated protocol grid (isogeny G <= 4, bits G < t) is too shallow to locate an optimum. Extending the grid (isogeny G >= 5, which the code allows for G < t; bits is capped at G = t-1 by the `G < t` rule) would be a new protocol, not done here.
- The protocol's hybrid `guess2` (second summand) is not implemented.
- Per-relation figures rest on frozen lambda (from the ladder's cofactor-aware count) and on `coverage_exact`; no replay certificate, validator report or matched rho reference is attached, so under the validation gate nothing here is a leaderboard or `benchmarks/` row. The distance to the matched rho was not computed: this experiment compares arms of one oracle only (PROTOCOL.md), and native uncharged work (system specialisation, grid reading) is listed in the `native` field of each JSON line, not priced.
- Executed on the agent host, sequentially, not on EKS (see Measurement).

Coordinator review (main session, 2026-11-02)

- **Per-relation figures are modelled, not observed.** PR(λ) and PR(cov) divide measured F4 work per target by a frozen yield (λ or `coverage_exact`). The grid observed 30 hits in 2,700 targets in total, all verified. That count matches coverage and cannot itself rank cells.
- **The fall in F4 cost is not moved into uncharged work.** The uncharged native counters (`pkm_system_fp_mults_uncharged`, `pkm_grid_fp_mults_uncharged`) are below 0.05% of the charged F4 multiply-adds per target in every cell (largest share: isogeny t = 6, G = 4, 288 vs 7.1e5 per target). Pricing them at w = 0 / 0.1 / 1 does not change any ordering in this README.
- **What deep guessing approaches.** As G → t each branch system covers a sub-tower of 2^(t−G) elements, so the oracle tends towards enumeration of the base. The measured best cell (isogeny t = 6, G = 4) is 78× below the previous algebraic best in F4 multiply-adds per modelled relation. That is the improvement this lead measures. It is **not** compared with the generic pair-table baseline (`../curve25519_ic_baseline_20261102`) in a common unit, so no distance to rho is claimed. Converting multiply-adds to counted GAE and running both through ecbench is the next step.